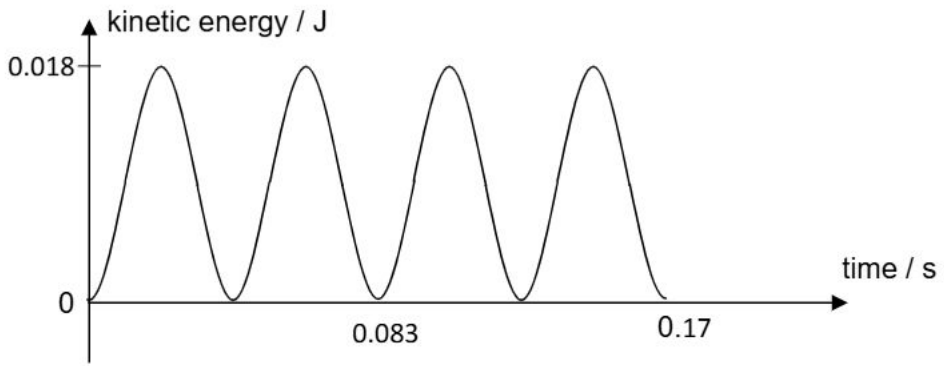
Question 3		
(a)	$\omega = 2\pi f = 2\pi(12) = 75.4 = 75 \text{ rad s}^{-1}$	**A1**
(b)(i)	$a_0 = \omega^2 x_0 = (75)^2 (0.016)$ $= 90 \text{ m s}^{-2}$	**C1** **A1**
(b)(ii)	Weight is in same direction as contact force at top, opposite to contact force at bottom. Same net force at both positions, hence the contact force at bottom is larger.	**B1** **B1**
(c)	$\frac{1}{2}mv_0^2 = \frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2}(0.025)(75)^2(0.016)^2 = 0.018 \text{ J}$ $T = \frac{1}{f} = \frac{1}{12} = 0.08333 \text{ s}$  **B1** – correct number of cycles **B1** – correct shape	**A1**